

Thinh Ong

Infectious disease modelling and vaccine epidemiology

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I build mathematical models of vaccine-preventable disease to strengthen outbreak surveillance and immunisation policy in Vietnam and the global south.

EDUCATION

2022-2026	PhD , University of Oxford, United Kingdom
2012-2018	MD , University of Medicine and Pharmacy at Ho Chi Minh City, Vietnam

EXPERIENCE

2021-2022	Research Assistant , Mathematical Modelling Group, Oxford University Clinical Research Unit, Vietnam
2020-2021	Editorial Assistant , MedPharmRes, University of Medicine and Pharmacy at Ho Chi Minh City, Vietnam
2019-2020	Research Intern , Cancer Epigenetics Lab, Garvan Institute of Medical Research, Australia
2018-2023	Biostatistical Consultant , Infertility Department, Tu Du Hospital, Vietnam

GRANTS

Secured \$237,930 in research grants since 2022.

2025-2026	<i>PI</i>	\$55,456	Understanding how immunisation inequities drive measles outbreaks. University of Oxford, Improving Equitable Access to Healthcare pump priming.
2024-2025	<i>PI</i>	\$27,694	Modelling measles and hand-foot-mouth disease transmission dynamics in Ho Chi Minh City. OUCRU Programme Scientific Committee.
2024-2025	<i>Co-I</i>	\$104,382	Research cultures in Southeast Asia. Wellcome.

2023-2025	PI	\$29,961	Impact of a 15-month disruption in childhood vaccination on coverage and seroprevalence in Ho Chi Minh City. OUCRU Programme Scientific Committee.
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A W A R D S

2022-2026	PhD Studentship , OUCRU and NDM Tropical Network Fund, University of Oxford		
2019	Outstanding Presentation , 36th Science and Technology Conference, Ho Chi Minh City		
2018	Highest GPA in the 6-year MD programme, the highest in school history		
2018	Highest dissertation score		

R E S E A R C H T E A M

<i>TENURE</i>	<i>ROLE</i>	<i>NAME</i>	<i>NEXT DESTINATION</i>
2024-2026	RA	Pham Nguyen The Nguyen	MPH, National Cancer Center Graduate School of Cancer Science and Policy
2023-2026	RA	Phan Truong Quynh Anh	PhD, University of Oxford
2022-2023	RA	Huynh Ngoc Tuyen	PhD, University of Oxford
2024	Intern	Nguyen Thanh Lam	MPH, University of Caen Normandie
2024	Intern	Julian Hough	BSc, Princeton University
2024	Intern	Huynh Xuan Loc	PhD, Boston University

P O L I C Y E N G A G E M E N T

2024-2025	Evaluated the city-wide infectious disease surveillance system
2024-2025	Led the benchmarking and selection of algorithms for defining epidemic thresholds
2024	Developed a surveillance dashboard tracking time-varying reproduction numbers to support the CDC response
2024	Authored a rapid measles seroprevalence report to advise the Department of Health during an active outbreak

TEACHING

YEAR	PEOPLE	COURSE
2025	30	Modelling and Analysis of Serological Data. Teaching assistant for a short course by MRC GIDA, Imperial College London and NUS. OUCRU.
2024-2025	15	Infectious disease modelling for babies. Methods for mathematical models of infectious diseases. HCMC CDC.
2023	60	Introduction to mathematical models for controlling infectious diseases. Compartmental models and reproduction numbers. HCMC CDC.
2024	40	R for babies. Monthly practical R programming classes. HCMC CDC.

SOFTWARE

2025	denim. Deterministic discrete-time compartmental models with customisable dwell-time distributions. thinhong.github.io/denim
2025	serosv. Analyse serological data and estimate infectious disease parameters. oucru-modelling.github.io/serosv
2025	tvmedg. Causal mediation analysis using g-computation. causalepi.github.io/tvmedg
2021	MethPanel. Multiplex bisulphite PCR sequencing for DNA methylation biomarker panels. github.com/thinhong/MethPanel

SELECTED PUBLICATIONS

1. **Ong T**, Thuy CT, Tam NH, et al., Choisy M, Tan LV. Detection of immunity gap before measles outbreak, Ho Chi Minh City, Vietnam, 2024. *Emerging Infectious Diseases*. 2025;31(10). [doi](#)
2. **Ong T**, Phan ATQ, Ha LM, Choisy M. denim: an R package for deterministic compartmental models with flexible dwell-time distributions. *Methods in Ecology and Evolution*. 2026;17(4):1060-1068. [doi](#)
3. Nguyen QD, **Ong T**, Truong LTT, Trinh TH, Le HN, Do LAH, Huynh J. High vulnerability of infants under 6 months of age in Vietnam's measles outbreak. *Vaccine*. 2026;77:128375. [doi](#)
4. Leung K, Cook AR, Wu JT, et al., **Ong T**, et al., Dabak SV. Enhancing global health security responses through greater inclusion of the global south in infectious disease modelling. *BMJ Global Health*. 2025;10(8). [doi](#)
5. Phi HD, Ca TN, Oanh PM, Phong NT, **Ong T**, Nhung PTH. Impact of measles vaccination on disease severity and economic burden among children in northern Vietnam, 2017-2019. *Vaccine: X*. 2025;26:100714. [doi](#)

6. Tran DM, **Ong T**, Cao TV, Pham QT, Do H, Phan PH, Choisy M, Pham NTH. Hospital-acquired infections and unvaccinated children: the 2017-2019 measles outbreak in northern Vietnam. *BMC Infectious Diseases*. 2024;24(1):948. [doi](#)
7. Long KQ, Ngoc-Anh HT, Phuong NH, et al., **Ong T**, et al., Minh HV. Clustering lifestyle risk behaviors among Vietnamese adolescents: a Bayesian multilevel analysis. *The Lancet Regional Health, Western Pacific*. 2021;15:100225. [doi](#)
8. Luu PL, **Ong T**, Loc TTH, Lam D, Pidsley R, Stirzaker C, Clark SJ. MethPanel: a parallel pipeline for multiplex bisulphite PCR sequencing. *Bioinformatics*. 2021;37(15):2198-2200. [doi](#)
and 19 more publications.